μPC154

Instrumentation Operational Amplifier

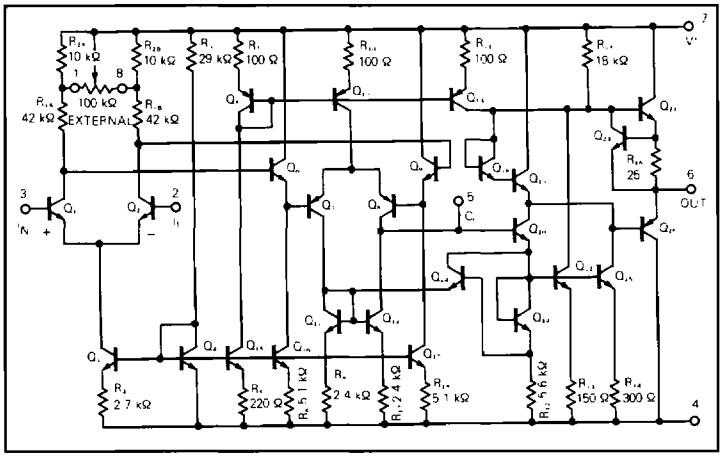
GENERAL DESCRIPTION

The μPC154 is an operational amplifier featuring superior performance in applications where low noise, low drift and accurate closed-loop gain are required. High common mode rejection and offset null capability make this device suitable for low level instrumentation applications over a wide supply voltage range.

FEATURES

- High Open Loop Gain — 3000 000 Typ.
- Low Input Voltage drift — 0.5 μV/°C typ.
- Low Input Offset Current — 3 nA Typ.
- High Common Mode Rejection — 120 dB Typ.
- High Input Voltage Range — ±14 V Typ.
- Wide Power Supply Range — ±3 V to ±22 V
- Offset Null Capability
- μA725 Direct Replacement

EQUIVALENT CIRCUIT



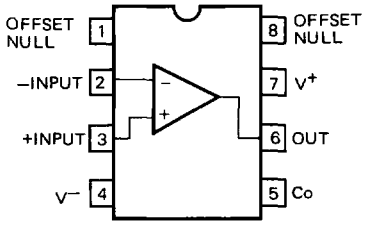
ORDERING INFORMATION

μPC154D



8 pin Ceramic DIP
(Dual In-Line Package)

CONNECTION DIAGRAM (Top View)



μPC154D

ABSOLUTE MAXIMUM RATINGS (Ta = 25°C)

PARAMETER			μPC154	UNIT
Voltage between V^+ and V^-			44	V
Power Dissipation*	D	Package	500	mW
Differential Input Voltage			±5	V
Input Voltage (Note 1)			±22	V
Voltage between Offset-Null and V^-			±0.5	V
Operating Temperature Range	D	Package	−20 to +80	°C
Storage Temperature Range	D	Package	−55 to +150	°C

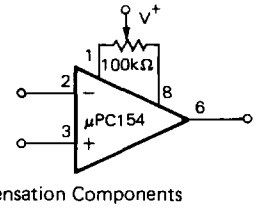
Note 1: For supply voltages less than ±15 V, the absolute maximum input voltage is equal to the supply voltage.

* See thermal information in chapter 11.

ELECTRICAL CHARACTERISTICS (Ta = 25°C, $V^{\pm}$ = ±15 V)

CHARACTERISTIC	MIN.	TYP.	MAX.	UNIT	CONDITIONS
Input Offset Voltage		0.5	2.5	mV	Rs ⩽ 10 kΩ
Average Input Offset Voltage Drift		0.5	3.0	μV/°C	Rs⩽50 Ω (With external trim)
Average Input Offset Voltage Drift		2.0		μV/°C	Rs⩽50 Ω (Without external trim)
Input Offset Current		3.0	35	nA	
Average Input Offset Current Drift		10		pA/°C	
Input Bias Current		50	125	nA	
Input Impedance		1.5		MΩ	
Large Signal Voltage Gain	250,000	3,000,000		–	R_L ⩾ 2 kΩ, Vo = ±10 V
Power Consumption		80	150	mW	
Output Voltage Swing	±12	±13		V	R_L ⩾ 10 kΩ
Output Voltage Swing	±10	±13		V	R_L ⩾ 2 kΩ
Common-Mode Input Voltage Range	±13.5	±14		V	
Common-Mode Rejection Ratio	96	120		dB	Rs ⩽ 10 kΩ
Supply Voltage Rejection Ratio		2.0	35	μV/V	Rs ⩽ 10 kΩ
Input Noise Voltage		2		μV_{p-p}	Rs = 10 kΩ f = 0.1 ~ 10 Hz
Input Noise Voltage		0.3		μV rms	Rs = 10 kΩ f = 10 Hz ~ 1 kHz

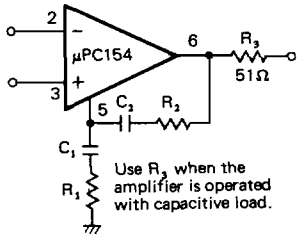
Offset Voltage Null Circuit



Frequency Compensation Components

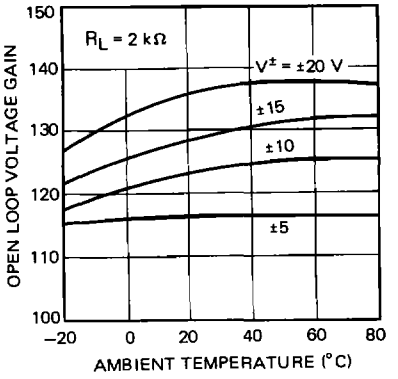
GAIN	R_1 (Ω)	C_1 (μF)	R_2 (Ω)	C_2 (μF)
10,000	10 k	50 pF	–	–
1,000	470	0.001	–	–
100	47	0.01	–	–
10	27	0.05	270	0.0015
1	10	0.05	39	0.02

Frequency Compensation Circuit

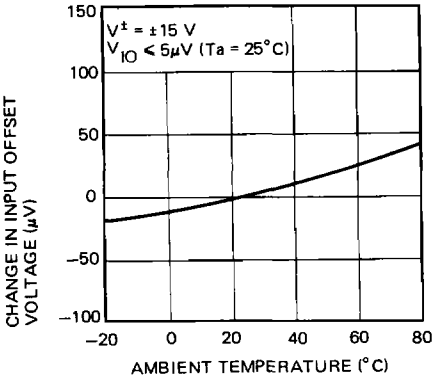


TYPICAL PERFORMANCE CHARACTERISTICS (Ta = 25°C)

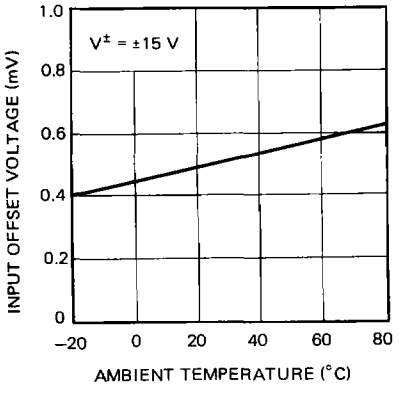
OPEN LOOP VOLTAGE GAIN



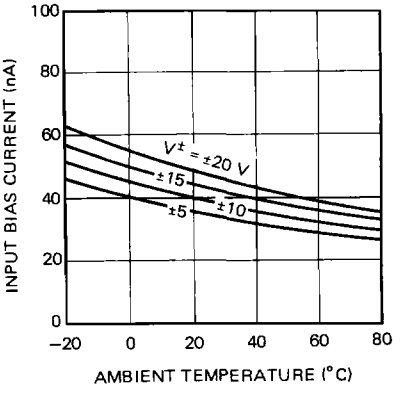
NULLED INPUT OFFSET VOLTAGE



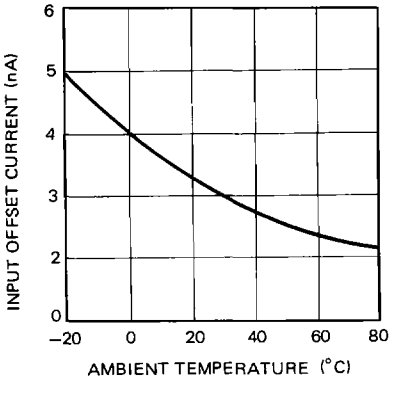
UNNULLED INPUT OFFSET VOLTAGE



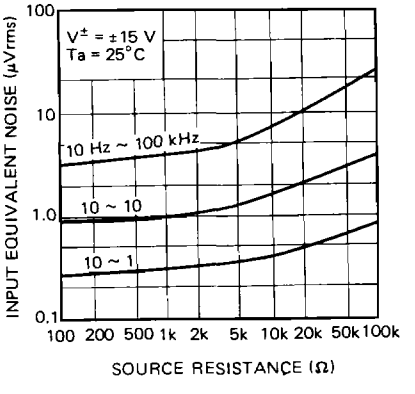
INPUT BIAS CURRENT



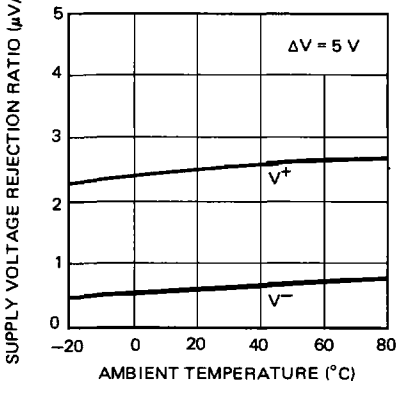
INPUT OFFSET CURRENT



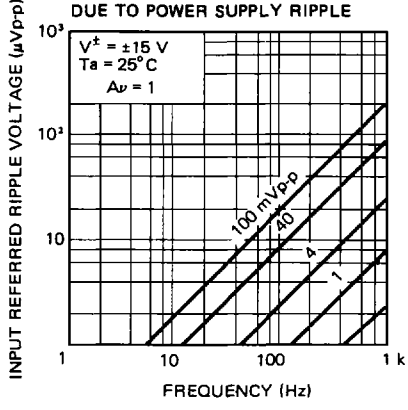
INPUT EQUIVALENT NOISE



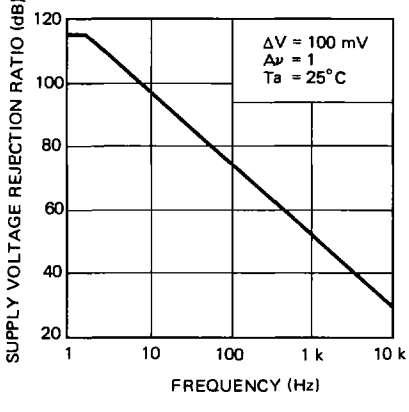
SUPPLY VOLTAGE REJECTION RATIO



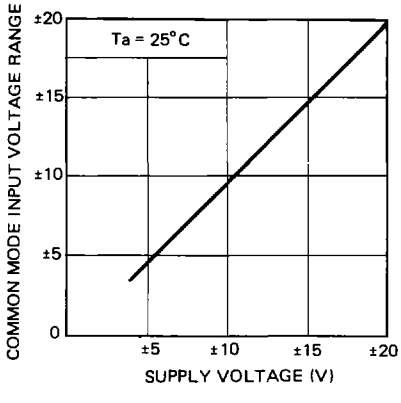
EQUIVALENT INPUT RIPPLE VOLTAGE DUE TO POWER SUPPLY RIPPLE



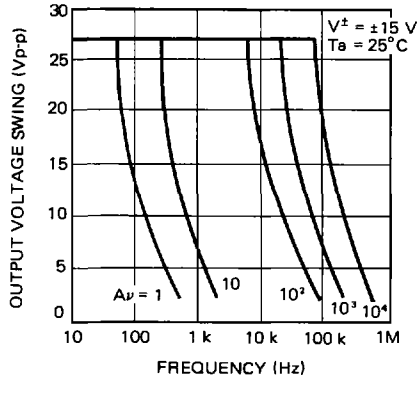
SUPPLY VOLTAGE REJECTION RATIO



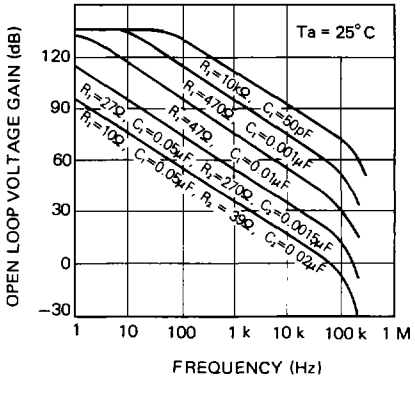
COMMON MODE INPUT VOLTAGE RANGE



LARGE SIGNAL FREQUENCY RESPONSE FOR RECOMMENDED COMPENSATION NETWORKS

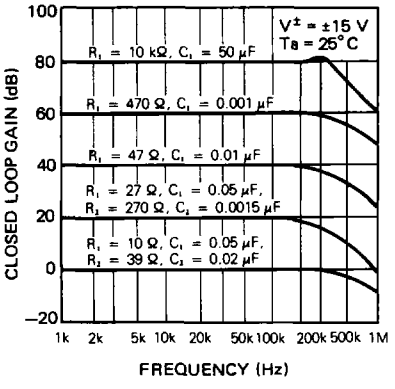


OPEN LOOP FREQUENCY RESPONSE USING RECOMMENDED COMPENSATION NETWORKS

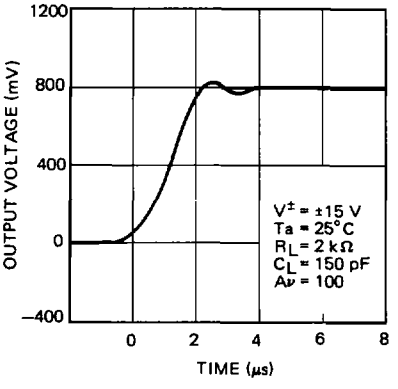


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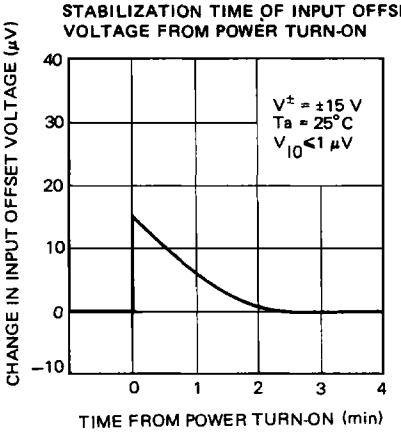
FREQUENCY RESPONSE FOR VARIOUS CLOSED-LOOP GAINS



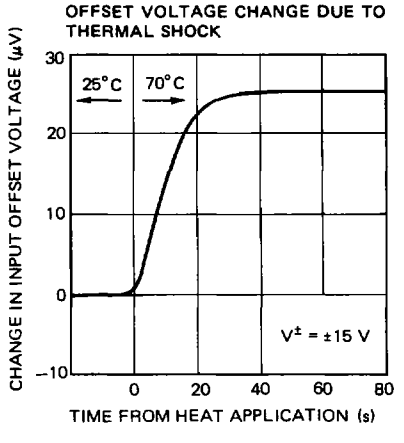
TRANSIENT RESPONSE



STABILIZATION TIME OF INPUT OFFSET VOLTAGE FROM POWER TURN-ON



OFFSET VOLTAGE CHANGE DUE TO THERMAL SHOCK



POWER CONSUMPTION

